Muhammad Raheel Anwar

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NUST, Islamabad, Pakistan

Objective

Seeking a challenging Senior AI Consultant position to apply my expertise in **Agentic AI Systems, Multi-Agent Frameworks, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Enterprise AI Solutions**. Experienced in designing, developing, and deploying production-grade AI systems, intelligent automation platforms, and scalable machine learning pipelines. Proven ability to translate cutting-edge AI technologies into real-world applications across banking, telecommunications, and enterprise domains, delivering measurable business impact through automation, decision intelligence, and AI-driven innovation.

Interests

Agentic AI Systems · Multi-Agent Frameworks · Autonomous AI Systems · AI Agent Orchestration · Financial AI · PySpark · Expert SQL · Banking Automation · Decision Intelligence Systems · Intelligent Process Automation · Enterprise AI Solutions · Knowledge Graphs · Microsoft Fabric · Azure · AWS · Power BI · AI Governance · MLOps & LLMOps · Low-Resource NLP · AI Code Intelligence · Recursive Self Improvement (RSI)

Education

- **National University of Sciences & Technology (NUST)** *Oct 2022 – July 2025*
  *M.Sc. in Computer Science - **GPA**: 3.70/4.00* Islamabad, Pakistan
  - Thesis: *"MCQs Generation Using Large Language Models: A Mixture-of-Agent Approach" with state-of-the-art results surpassing flagship models*
  - Relevant Coursework: Generative AI, Deep Learning, Machine Learning, Blockchain Technologies
- **NFC Institute of Engineering & Technology (NFC IET)** *Oct 2016 – Sept 2020*
  *B.Sc. in Computer Science - **GPA**: 3.49/4.00* Multan, Pakistan
  - Projects: *Believers Game* and *Dynamic Shopping App*
  - Relevant Coursework: Object-Oriented Programming, Algorithms, Operating Systems

Experience

- **Systems Limited** *Nov 2025 – Present*
  Senior Consultant AI Islamabad, Pakistan
  - Providing enterprise AI consultancy and developing cutting-edge AI solutions for global financial and telecommunication institutions across the Middle East.
  - **Mashreq Bank UAE (Dubai, UAE) — External Consultant AI | May 2026 – Present**
    - Led two cross-functional engineering teams driving enterprise AI, process analytics, and intelligent automation initiatives.
    - **Process Analysis & Data Analytics Dashboards:** Developed comprehensive process analysis dashboards tracking end-to-end customer/operational journeys with automated suggestions, findings, and deep data analytics extensions providing rich insights and recommendations to higher management.
    - **Departmental E2E Intelligence Dashboards:** Built end-to-end dashboards across diverse internal departments integrating AI-driven recommendations, anomaly observations, predictive analytics, and trend forecasting.
    - **Internal Chatbots & Automation:** Designed and deployed custom internal chatbots and intelligent workflow automations to streamline internal operations and enhance enterprise productivity.
  - **McKinsey & Company (KSA) (Riyadh, Saudi Arabia) — External Consultant AI | Feb 2026 – May 2026**
    - Contributed to the development of AI-powered banking solutions for financial institutions across the Middle East.
    - Worked closely with domain experts and engineering teams to integrate predictive analytics into banking workflows.
    - **Loan Prediction & Risk Assessment:** Developed machine learning and LLM-powered solutions for loan eligibility prediction, credit analysis, and decision support.
    - **AI-Driven Automation:** Automated data processing and business workflows to improve operational efficiency and accelerate decision-making processes.
  - **Ooredoo Qatar (Doha, Qatar) — External Consultant AI | Nov 2025 – Feb 2026**
    - Developed enterprise-grade AI solutions leveraging LLMs, RAG architectures, and intelligent automation for telecom operations.
    - Collaborated with business and technical stakeholders to identify automation opportunities and improve customer engagement workflows.

    - **Enterprise Chatbots & Knowledge Systems:** Built scalable RAG-based chatbots using Azure AI Foundry and LangChain for enterprise knowledge retrieval and customer support.
    - **Process Automation:** Designed AI-driven automation workflows to streamline operational processes and reduce manual intervention across support functions.
  - **MLOps & Infrastructure:** Implementing end-to-end MLOps pipelines on Databricks for model lifecycle management, ensuring reproducible training and deployment via Docker and Kubernetes.

- **Intelligent Robotics Lab, NCAI, NUST** *Mar 2025 – Nov 2025*
  Senior Research Officer - R&D Islamabad, Pakistan
  - Led research in Agentic AI and RAG-based systems, including VLM-based detection (FastVLM) and plant disease detection.
  - **Foundational Urdu LLM:** Led the development of a custom Urdu Large Language Model, including tokenizer training, distributed pre-training on high-performance computing clusters, and fine-tuning for downstream tasks.
  - **Autonomous Coding Agent:** Engineered a specialized agentic framework architecture for codebase manipulation, implementing Model Context Protocol (MCP) for seamless tool integration and designing multi-agent architectures that automate refactoring and feature implementation.
  - **AI-Code Security & Test Automation:** Developed an automated verification pipeline for AI-generated code, utilizing static and dynamic analysis to detect vulnerabilities and ensuring production-readiness through rigorous security-focused test generation.
  - **Juntendo University Medical System:** Architected a comprehensive medical advice system integrating VLM-based diagnostics and LLM-powered patient interaction, deployed as a full-stack mobile solution.
  - **FastVLM & Plant Disease Detection:** Pioneered research in VLM localization (FastVLM) and developed a plant disease detection system achieving 28% higher accuracy than traditional CNNs.
  - **Custom Voice Agents:** Developed real-time, low-latency conversational voice agents for university help desks, integrating GPT-4, ElevenLabs, and VAPI.

- **IZBA USA** *Aug 2020 – Dec 2024*
  AI Consultant USA - Remote
  - **Autonomous Data Agents:** Designed and deployed intelligent scraping, extraction, and verification agents to autonomously collect, validate, and enrich logistics data from diverse web sources and enterprise platforms.
  - **Supply Chain Predictive Modeling:** Developed machine learning models for missing value prediction, demand forecasting, carrier performance estimation, and trend analysis across 3PL and 4PL logistics providers.
  - **Data Quality & Entity Resolution:** Built automated data cleaning, deduplication, and entity-matching pipelines to improve dataset consistency and reliability across millions of logistics records.
  - **Knowledge Graphs & Data Integration:** Integrated heterogeneous logistics datasets from multiple sources, constructing unified data repositories for downstream analytics and decision support.
  - **Large-Scale Web Intelligence:** Engineered scalable data acquisition pipelines for monitoring transportation networks, freight providers, warehouses, and supply chain ecosystems in real time.
  - **MLOps & Automation:** Automated model training, evaluation, and deployment workflows, enabling continuous data processing and predictive analytics in production environments.
  - **Research & Experimentation:** Conducted extensive data preprocessing, feature engineering, exploratory analysis, and model benchmarking to support machine learning research and operational decision-making.

PUBLICATIONS

[S.1] Raheel Anwar et al. (2025). **MCQs Generation Using Large Language Models: A Review**. Manuscript accepted for publication in *IEEE Access*. Published: January 16, 2026. https://doi.org/10.1109/ACCESS.2026.3652006

[S.2] Raheel Anwar et al. (2025). **MCQsAgent: Towards Human-Level MCQs Generation via Collaborative Multi-Agent AI Framework**. Manuscript accepted for publication in *Expert Systems With Applications*. Published: May 26, 2026. https://doi.org/10.1016/j.eswa.2026.132876

[S.3] Raheel Anwar (2026) **The Productivity Myth and Learning Paradox of Generative AI in Higher Education**. Submitted to *International Journal of Educational Technology in Higher Education (IJETHE)*. https://doi.org/10.21203/rs.3.rs-9869630/v1

[S.4] Basit Ali, **Raheel Anwar** (2026) **Routing-level Corpus Poisoning in Stateful Agentic RAG Pipelines: Attack, Defenses, and the Limits of State-transition Monitoring**. https://dx.doi.org/10.2139/ssrn.7199198

[S.5] Basit Ali, **Raheel Anwar** (2026) **Signed but Unsafe: Securing Runtime Tool Selection in LLM Agents under Adversarial Metadata Injection**. https://dx.doi.org/10.2139/ssrn.7199341

[S.6] Aqsa Arif, **Muhammad Raheel Anwar** (2026) **Machine Learning-Guided Cleaner Production of Bio-Dyed Silk via Floral Waste Valorization**. https://dx.doi.org/10.2139/ssrn.7494639

Skills

- **AI & Machine Learning:** Agentic AI, LLMs, RAG, Finetuning, Deep Learning, Computer Vision, NLP, Multi-Agent Systems
- **Languages & Frameworks:** Python, TensorFlow, PyTorch, LangChain, OpenCV, Scikit-learn, C++, Dart, Java
- **MLOps & Cloud:** MLflow, Docker, Kubernetes, AWS, Azure, Databricks, AI Foundry, CI/CD
- **Web & Mobile:** FastAPI, Flask, Flutter (Integration), REST APIs
- **Databases:** PostgreSQL, MySQL, MongoDB, VectorDB, ElasticSearch
- **Research Skills:** Systematic Literature Review, Academic Writing, Experimental Design, Data Analysis

Awards and Recognitions

- **Nominated for Best Methodology Award - MS Thesis** *2025*
  Graduate Studies Committee
  ◦ Nominated for the Best Methodology Award for innovative approach in MS thesis research.
- **Best Duo Final Year Project - Dynamic Adaptive UI App** *2020*
  NFC Institute of Engineering & Technology (NFCIET), Multan
  ◦ Developed a dynamic and adaptive UI mobile app that can be converted into multiple domains like grocery, pharmacy, etc.
- **Finalist Winner, ICE 2019 - Deev: Ride Sharing App** *2019*
  Innovation and Creativity Event (ICE), NFCIET Multan
  ◦ Finalist winner with "Deev", a ride-sharing mobile app built in Flutter.
- **Runner-up, ICE 2018 - Anonymous Web Browser** *2018*
  Innovation and Creativity Event (ICE), NFCIET Multan
  ◦ Developed an anonymous and lightweight web browser as part of the Innovation and Creativity Event.

Certifications

- **Google Prompting Essentials Specialization – Google** *July 2026*
- **Machine Learning Specialization – Stanford University & DeepLearning.AI** *July 2026*
- **Credly Certifications (12 Total)**
- **What is Data Science? – IBM**
- **Data Science Methodology – IBM**
- **Microsoft Azure Fundamentals (AZ-900) – Microsoft Press and Jim Cheshire**
- **Building RAG Solutions with Azure AI Foundry – Ziggy Z**

Additional Information

Languages: English (Fluent), Urdu (Native), Punjabi (Intermediate), Arabic (Beginner)
Interests: Gaming, Singing, Rapping, History, Exploring new tools in tech

References

1. **Dr. Shah Khalid**
   Assistant Professor, Department of Computing
   School of Electrical Engineering and Computer Science (SEECS)
   National University of Sciences and Technology
   H – 12 Islamabad, Pakistan
   Email: shah.khalid@seecs.edu.pk
   Relationship: Thesis Advisor
2. **Hafiz Syed Muhammad Bilal**
   Assistant Professor, Department of Computing
   School of Electrical Engineering and Computer Science (SEECS)
   National University of Sciences and Technology (NUST)
   H – 12 Islamabad, Pakistan
   Email: bilal.ali@seecs.edu.pk
   Relationship: GC Member